

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- a. Participant Name: Anshul Jain (Team BloodWyrn, Solo)
- b. Problem Statement: AI for Better Living and Smarter Communities.
Create a data intelligence tool people would actually use, and show how acceleration helps them make a faster or better decision.

Brief about the idea

VayuSense is an AI **Decision Intelligence Platform** that turns millions of raw air quality sensor readings across Indian and APAC cities into one honest, decision-ready answer, for parents, schools, clinics, and city officials.

It pairs **NVIDIA RAPIDS** accelerated data processing (**37.5× faster** than pandas over **5.9M+ real OpenAQ sensor readings**) with a **Google ADK** multi-agent pipeline (**Data Analyst → Health Advisor**, powered by **Gemini**) to go beyond a raw AQI number: surfacing a **Safety Score** and a **Human Impact** metric (cigarette-equivalent exposure, illustrative life-expectancy impact) that most air quality tools never show.

Built solo, fully deployed, and designed to scale from a hackathon prototype into a real public-health decision tool for any city.

Your solution should be able to explain the following:

- How did you approach the common problem statement and translate it into a working solution using Google Cloud technologies (e.g., BigQuery, ADK, Vertex AI, Gemini, etc.) while ensuring the hackathon's core requirements are met? If you selected the NVIDIA problem statement, also explain how NVIDIA technologies were integrated into your solution.
- What real-world problem does your solution address, and what practical impact does it create for users, organizations, or communities??
- What is the core architecture or workflow behind your solution, and how does it transform data into actionable insights or intelligent decisions?

Approach: Ingested **5.9M+ real OpenAQ sensor readings**; built a GPU-accelerated (**NVIDIA cuDF/RAPIDS**) pipeline for cleaning, trend and anomaly analysis; layered a **Google ADK Sequential multi-agent system** (Data Analyst → Health Advisor) on **Gemini** to reason over the results; shipped as a publicly deployed FastAPI app.

Real-world impact: Delhi-NCR breathes the world's most hazardous urban air. **VayuSense** turns that into a same-day decision for schools (sports practice), clinics (staffing), and city wards (advisories); not just another chart nobody reads.

Architecture: OpenAQ archive → NVIDIA RAPIDS cuDF (GPU) → Google ADK/Gemini agents → FastAPI dashboard → end users. Converting raw sensor data into a **Safety Score**, a **Human Impact** estimate, and a plain-language, decision-ready recommendation.

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

Proven acceleration, not a claim. An honest, reproducible **37.5× cuDF-vs-pandas benchmark** over 5.9M+ real rows; most entries will show a number with no receipts.

Human stakes, not just AQI. Cigarette-equivalent exposure and an illustrative life-expectancy impact, grounded in published air quality epidemiology (Berkeley Earth, AQLI-style); a stat most air quality tools never surface.

A true multi-agent pipeline, not a prompt wrapper. Google ADK Sequential agents: the **Data Analyst** gathers facts with real tools; the **Health Advisor** reasons over them.

Answers the brief end to end. A real ingest, clean, model, visualize pipeline; 2+ Google Cloud layer items; 2+ NVIDIA layer items; a genuinely useful output.

Built to scale beyond the hackathon. The same architecture works for any city or pollutant dataset; just repoint the ingestion.

List of features offered by the solution

Live multi-city dashboard (Delhi, Mumbai, Kolkata, Chennai) with 90-day trend and 7-day rolling average per pollutant

Safety Score (0 to 100) with a plain-language verdict for each city

Human Impact panel: cigarette-equivalent exposure and estimated life-expectancy impact

Pollution hotspot ranking of the worst-performing stations per city

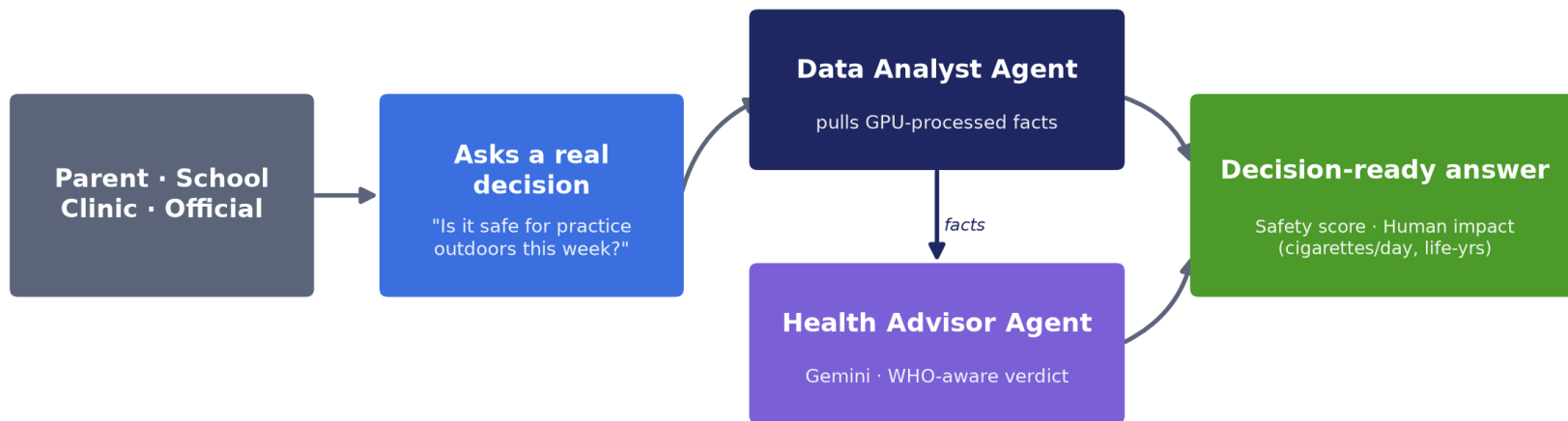
“Ask VayuSense” natural-language chat, a **Google ADK multi-agent pipeline** (Data Analyst → Health Advisor) on Gemini

GPU-accelerated backend analytics (NVIDIA RAPIDS cuDF) with a live benchmark panel

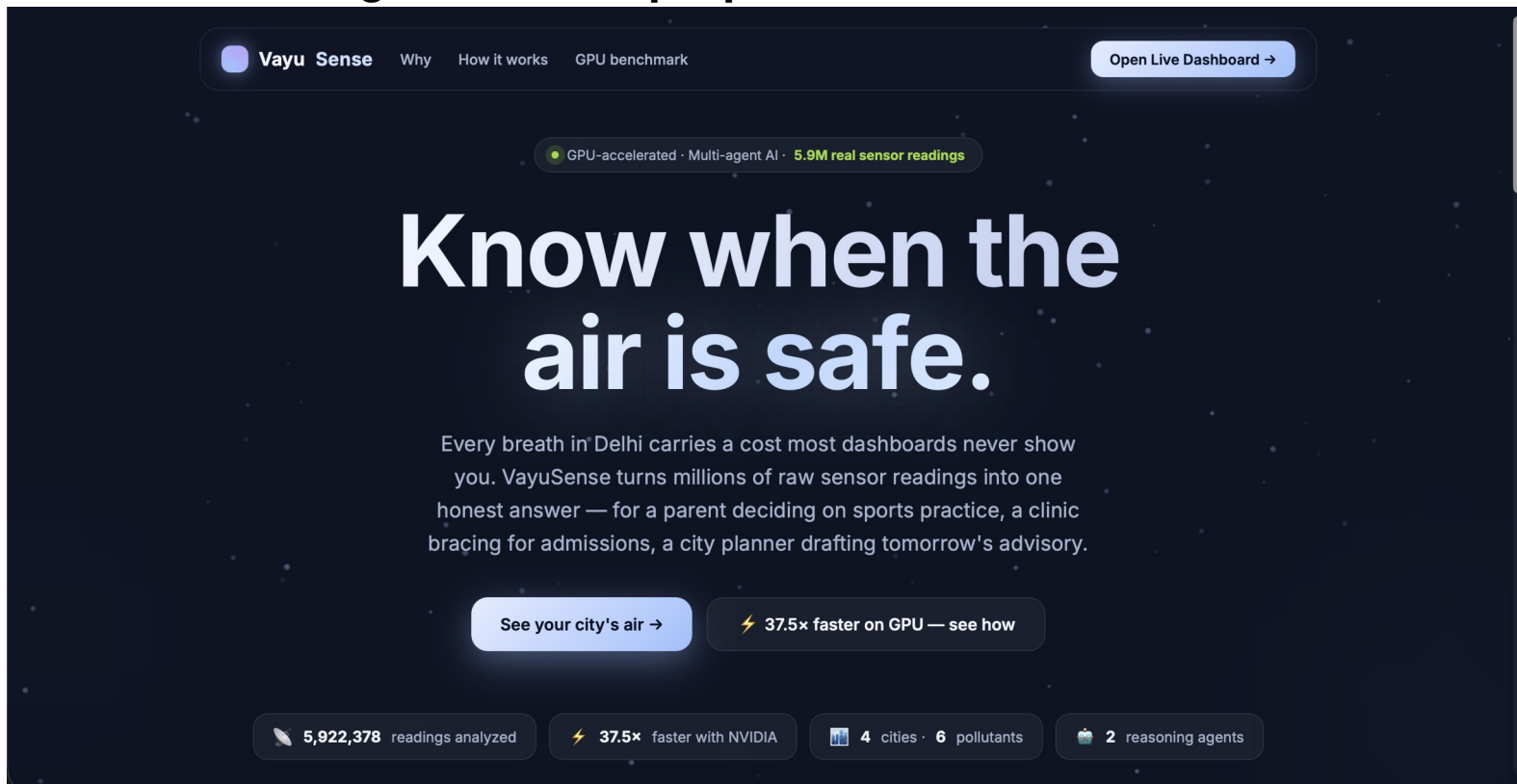
Fully deployed, publicly accessible web app with WHO-guideline comparisons across 6 pollutants

Process flow diagram or Use-case diagram

VayuSense: Sequential Multi-Agent Pipeline (Google ADK + Gemini)

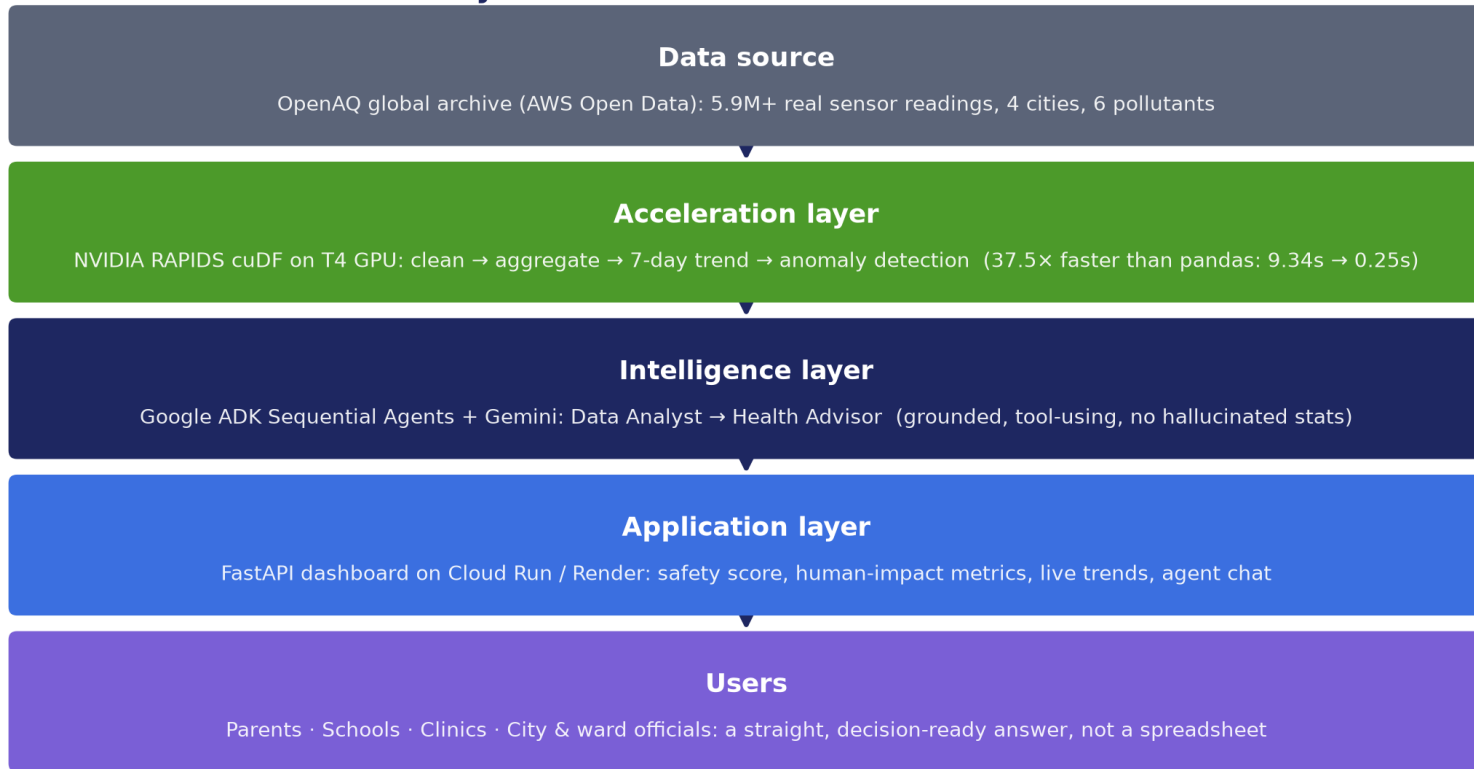


Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:

VayuSense: End-to-End Architecture



Technologies / Google / Nvidia Services used in the solution

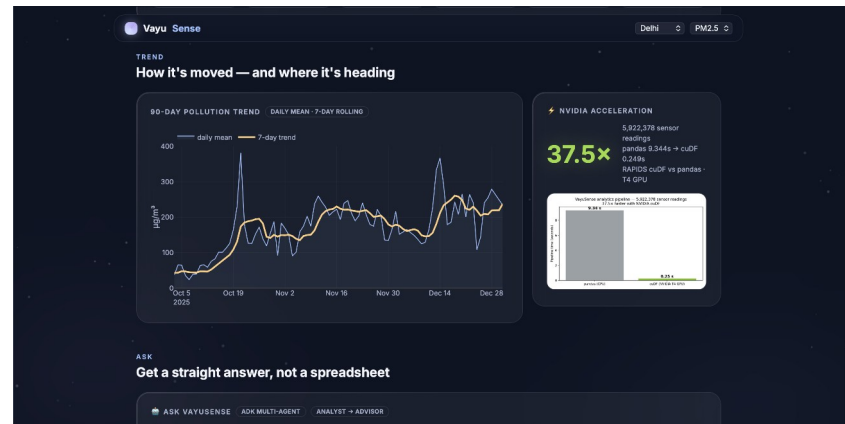
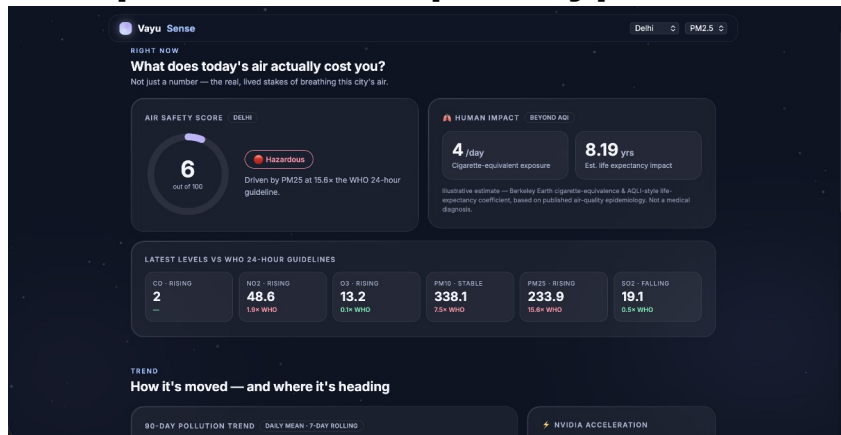
(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Google stack: **Google ADK** (Agent Development Kit) for Sequential multi-agent orchestration; **Gemini 2.5 Flash** for both agents' reasoning; a BigQuery-ready data pipeline; a containerized FastAPI app built for Cloud Run deployment.

NVIDIA stack: **NVIDIA RAPIDS cuDF** for the GPU-accelerated clean, aggregate, trend, anomaly pipeline; **NVIDIA T4 GPU** (Google Colab) as the benchmark and production-scale processing target.

Why this stack: millions of sensor rows need cuDF-class speed to stay decision-fresh; grounded, tool-calling ADK agents avoid hallucinated health claims that a single freeform prompt risks; a lightweight containerized FastAPI service scales horizontally and deploys anywhere (Cloud Run, Render, or on-prem) with no code changes.

Snapshots of the prototype





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Thank you

